Are Anterior Cruciate Ligament–reconstructed Athletes More Vulnerable to Fatigue than Uninjured Athletes?

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ABSTRACT

SMEETS, A., J. VANRENTERGHEM, F. STAES, H. VANDENNEUCKER, S. CLAES, and S. VERSCHUEREN. Are Anterior Cruciate Ligament–reconstructed Athletes More Vulnerable to Fatigue than Uninjured Athletes? *Med. Sci. Sports Exerc.*, Vol. 52, No. 2, pp. 345–353, 2020. **Introduction:** Fatigue has a negative impact on lower extremity neuromuscular and biomechanical control. Because anterior cruciate ligament reconstruction (ACLR) athletes show already neuromuscular/biomechanical deficits in an unfatigued state, the negative impact of fatigue may magnify these deficits or help expose other deficits. So far, this has only scarcely been assessed warranting further research. **Methods:** Twenty-one athletes who had an ACLR and 21 uninjured controls performed five unilateral landing tasks before and after a match simulation protocol, whereas muscle activation (vastus medialis, vastus lateralis, hamstrings medialis, hamstrings lateralis, gastrocnemius medialis, gastrocnemius lateralis, gluteus medius) and landing kinematics and kinetics of the hip, knee, and ankle joint were recorded. A two-way ANOVA with a mixed-model design (main effects for group and fatigue) was used to compare landing kinematics, kinetics, and muscle activation between groups, and prefatigue and postfatigue. To avoid unjustified reduction of the data to discrete values, we used one-dimensional Statistical Parametric Mapping. **Results:** Only two interaction effects were found: an increased postfatigue knee abduction moment and an increased postfatigue thorax flexion angle was found in the ACL injured legs but not in the uninjured legs of the ACL group or in the control group, during the lateral hop and the vertical hop with 90° medial rotation, respectively. **Conclusions:** This study showed that overall ACLR athletes and uninjured athletes have similar biomechanical and neuromuscular responses to fatigue. For two biomechanical parameters, however, we did find an interaction effect, suggesting that landing deficits in ACLR athletes may become clearer in certain tasks when fatigued. **Key Words:** ACL RECONSTRUCTION, FATIGUE, BIOMECHANICAL ALTERATIONS, NEUROMUSCULAR ALTERATIONS, RETURN-TO-SPORT

When athletes return to sport after an anterior cruciate ligament reconstruction (ACLR), they continue to demonstrate prolonged neuromuscular and biomechanical alterations (1,2). Those persistent deficits are expected to predispose those athletes for reinjury and posttraumatic knee osteoarthritis (3). Since biomechanical and neuromuscular patterns are modifiable (4), it seems crucial to identify the prolonged deficits to target postoperative rehabilitation and return-to-sport (RTS) criteria accordingly to ultimately result in a reduction of the long-term negative consequences associated with anterior cruciate ligament (ACL) injuries.

Over the last few years, increasing attention has been paid to the potential negative impact of fatigue on lower extremity neuromuscular (5–7) and biomechanical (8–10) control in uninjured athletes. A recent systematic review with meta-analysis (11) concluded that athletes have a more extended landing pattern when athletes are fatigued compared with before fatigue. Such extended landing pattern is associated with increased injury risk (12). Because no consistent changes in muscle activation or frontal plane kinematics and/or kinetics were found postfatigue, the impact of fatigue on neuromuscular and biomechanical control might be limited in uninjured athletes (13). However, as ACLR athletes show already neuromuscular and biomechanical deficits in an unfatigued state (1,2) and because of their lengthy rehabilitation and sports absence, one might expect that ACLR athletes might be more vulnerable to general fatigue effects than their uninjured peers and that fatigue might exaggerate the deficits that can already be present prefatigue. As such, fatigue might thus even further increase reinjury risk in ACLR athletes, for example at the end of match play.

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Furthermore, fatigue might not only exaggerate prefatigue deficits but may also reveal other deficits that only appear when biomechanical and neuromuscular control is further compromised. In this respect, testing in a fatigued state might improve testing sensitivity when evaluating neuromuscular and biomechanical control in ACLR athletes that return to sport.

The effect of fatigue on neuromuscular and biomechanical control in ACLR athletes during single-leg landing tasks has been scarcely assessed (7,14–17). So far, existing studies could not prove that ACLR athletes are more vulnerable to fatigue than uninjured athletes as those studies found similar postfatigue alterations in controls and ACLR athletes compared to prefatigue. However, the existing studies have several methodological limitations warranting further research. First of all, most studies assessed fatigue effects on landing patterns during relatively easy tasks, for example, during double (14) and single-leg (15) drop vertical jumps and single-leg drop landings (16). As Webster et al. (16) state in their limitation section, fatigue effects should also be investigated during more complex movements involving directional changes and tasks that require maximal performance as ACLR athletes might have enough reserve to cope with fatigue effects during easy tasks but not during highly demanding or more complex tasks (11). A second methodological shortcoming of the existing studies (7,14–16) is that kinematic, kinetic, and/or muscle activation data were reduced to discrete values, such as peak values or values at specific time points (often initial contact). With this approach, only alterations in peak amplitude or alterations at the preselected time points can be found. To avoid unnecessary and unjustified data reduction, Statistical Parametric Mapping (SPM) can be used which allows to compare entire landing profiles without data reduction. Finally, most existing studies solely focused on landing kinematics and kinetics, omitting fatigue effects on muscle activation, whereas muscle activation patterns seem to have an important role in ensuring knee joint stability and the development of posttraumatic knee osteoarthritis. For example, modeling studies showed that increased co-contraction of hamstrings and quadriceps can result in excessive compressive forces which might be detrimental for articular cartilage health (18,19).

Based on these limitations of existing studies, the aim of the present study was to comprehensively assess whether ACLR athletes are more vulnerable than uninjured athletes to fatigue during a set of challenging landing tasks that involve directional changes. We hypothesize that ACLR athletes will demonstrate more pronounced alterations in landing kinematic, kinetic and/or muscle activation profiles compared with controls after the performance of a protocol that simulates fatigue related with high-intensity multidirectional intermittent sports participation.

METHODS

Participants. Twenty-one athletes who have undergone an ACLR (semitendinosus autograft) were included in this study (see Table 1). Inclusion criteria were: men or women between 16 and 40 yr of age with a primary ACL injury, participation in a sport that involves cutting, pivoting and/or jumping at competitive level (minimum, two training days per week and one match per week) before their injury and the willingness to return to the same sport. Patients who had a history of a previous ACL injury or a previous serious knee injury were excluded. All patients completed rehabilitation with their own physiotherapist and were cleared by their surgeon and/or physiotherapist to fully participate in training sessions again. The testing took place maximum 2 wk before or after the first full training session. The average time postsurgery at the time of testing was 258.6 ± 54 d. Furthermore, a control group was included consisting of 21 uninjured athletes who were matched for age, gender and type of sport. The control subjects were free of lower extremity or back injuries for at least 6 months and had no history of ACL injuries.

TABLE 1. Participant characteristics.

	ACL	Control	*t*-test	
Participants (*n*)	21	21		
Height (cm)	178.8 ± 9.4	179.9 ± 9.4	$t = -0.375$	$P = 0.710$
Weight (kg)	76.4 ± 11.3	72.5 ± 9.3	$t = 1.255$	$P = 0.217$
Age (yr)	23.8 ± 4.2	21.5 ± 1.5	$t = 2.32$	$P = 0.028$
Gender	6 females/15 males	6 females/15 males		
Football	$n = 13$	$n = 13$		
Volleyball	$n = 4$	$n = 4$		
Basketball	$n = 2$	$n = 2$		
Hurdles	$n = 1$	$n = 1$		
Dancing	$n = 1$	$n = 1$		
Quadriceps strength[a] ($N \cdot m \cdot kg^{-1}$)			$t = 2.063$[b]	$P = 0.046$
Injured leg	2.56 ± 0.49		$t = 1.006$[c]	$P = 0.321$
Uninjured leg	2.75 ± 0.38	2.91 ± 0.55	$t = -2.421$[d]	$P = 0.026$
Hamstrings strength[a] ($N \cdot m \cdot kg^{-1}$)			$t = 1.964$[b]	$P = 0.057$
Injured leg	1.38 ± 0.32		$t = 0.591$[c]	$P = 0.558$
Uninjured leg	1.51 ± 0.28	1.56 ± 0.27	$t = -2.650$[d]	$P = 0.016$
LSI quadriceps (%)	92.9 ± 5.9	92.3 ± 14.7		
LSI hamstrings (%)	91.2 ± 14.1	92.4 ± 7.1		

[a]Isokinetic concentric strength normalized to body weight.
[b]Unpaired *t*-test comparing ACL injured leg vs control.
[c]Unpaired t-test comparing ACL uninjured leg vs control.
[d]Paired *t*-test comparing ACL injured vs uninjured leg.
n, number of participants; LSI, limb symmetry index (ACL: injured leg/uninjured leg, control: weaker leg/stronger leg).

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All participants wore standardized indoor footwear (Indoor Copa, Kelme, Elche, Spain). The participants signed a written informed consent and the study was approved by the local ethics committee with reference number S60182.

Protocol. The test session started with a standardized warm-up consisting of 5-min cycling on a stationary bike, 10 squats, and 10 squat-jumps. Subsequently, three isometric maximal voluntary contractions (MVC) of 5 s were performed for all muscle groups to normalize EMG data. The detailed MVC setup was described in a previous study of our group (20).

After the performance of the MVC, participants started the actual protocol which consisted of a set of landing tasks performed before and after a 5-min match play simulation protocol (SAFT5 protocol) (21). The SAFT5 protocol involves a shuttle run over a 15-m distance with changes of directions (e.g., running through a slalom, side stepping and backward running), changes in speed (e.g., alternation between jogging and sprinting), and several functional movements (e.g., agility ladder, counter movement jumps, and scissor jumps). Standardized verbal instructions via tape recording navigated participants throughout the protocol. The SAFT5 was proven to be valid to simulate peripheral and central match-related fatigue for at least 30 min after finishing the protocol (all subjects were able to perform the set of landing tasks in this time window) (21). The protocol of SAFT5 is described in detail elsewhere (21) and is a reduced version on the SAFT90 protocol which has been shown to have similar intensity as to match play.

The landing tasks included five different sets of unilateral tasks. An online tool (www.randomizer.org) was used for counterbalanced randomization between tasks. Each task was repeated until three valid trials were performed on both the dominant and nondominant leg. The dominant leg was defined as the preferred leg to kick a ball. For all five tasks, both take-off and landing were on the same leg and a trial was considered valid if the landing was central on the force plate and the subject could maintain their balance on one leg for 5 s after landing. Shuffling on the stance leg was not allowed. After familiarization, participants performed three valid trials of five unilateral tasks on the dominant and nondominant leg, according to the following instructions:

- *Single-leg hop for distance*: to jump as far as possible on one leg.
- *Medial & lateral hop*: to jump sideways over a 0.24-m high hurdle (0.015 m wide) on 1 leg, to cover a mediolateral distance that was half the leg length (i.e., distance between the anterior superior iliac spine and medial malleolus).
- *Vertical hop with 90° of medial or lateral rotation*: to jump as high as possible on one leg, while performing an inward/outward rotation of 90°.

Data collection. Three-dimensional kinematic data were collected using 10 MX-T20 optoelectronic cameras (VICON, Oxford, UK) sampling at 100 Hz, synchronized with GRF data recorded from two 46.4 × 50.8 cm OR6–7 force plates (AMTI, Watertown, USA) sampling at 1000 Hz. Each participant had 44 spherical reflective markers positioned according to the eight-segment "Liverpool John Moores University" model including feet, upper and lower legs, pelvis, and trunk. This model was previously described in detail and shown to be reliable for measuring kinematics and kinetics during drop vertical jumps (22).

A wireless EMG system, type: Cometa Mini Wave (Zerowire, Aurion, Milan, Italy) was used to record muscle activity at 1000 Hz of the vastus lateralis, vastus medialis, biceps femoris (referred to as lateral hamstrings [LH]) semitendinosus (referred to as medial hamstrings, MH), gluteus medius (GlutMed), gastrocnemius medialis, and gastrocnemius lateralis using surface electrodes positioned according to the SENIAM guidelines (23). All electrode locations were shaved and gently cleaned with 70% isopropyl alcohol to reduce skin impedance. Silver-silver chloride, pregelled bipolar surface EMG electrodes (Ambu Blue Sensor, Ballerup, Denmark) were placed over the muscle belly and aligned with the expected muscle fiber orientations, with 2-cm interelectrode distance.

All subjects had to score their perception of fatigue by giving a RPE of the legs (RPE-L) and breathlessness (RPE-B) using a 16-item BORG scale (24), both before and after the SAFT5 protocol. The prefatigue RPE scores were administered after finishing all landing tasks prefatigue meaning that some subjects experienced already some fatigue at that timepoint. We decided to use differential RPE to differentiate between cardiopulmonary exertion represented by RPE-B and neuromuscular fatigue represented by RPE-L (24). Besides the fact that RPE is a reliable, valid and easy to use proxy for exercise intensity, RPE also reflects training status with fitter athletes showing lower RPE values for a standardized task (25).

Data analysis. All modeling and data processing were undertaken in Visual 3D (v.6.01.07; C-Motion, Germantown, USA) and MATLAB (R2017a; The MathWorks, Nattick, MA). Marker trajectories and forces were filtered using a fourth-order low pass Butterworth filter with a cutoff frequency of 18 Hz. Subsequently hip, knee and ankle kinematics were calculated using a Cardan sequence of rotations and kinetics were calculated using inverse dynamics. We reported external joint moments in this study (e.g., an external knee flexion moment will flex the knee). Initial contact events were created when the vertical ground reaction force crossed a 10 N threshold. Only the data of the landing were analyzed, from initial contact until 500 ms after initial contact.

All raw EMG signals of the motion trials and of MVC trials were bandpass filtered (6–240 Hz), rectified and low pass filtered with a fourth-order zero-lag Butterworth filter at a cutoff frequency of 15 Hz. Subsequently, the filtered EMG signals of the landing tasks were normalized to the peak value obtained during the three isometric maximal voluntary contractions.

In the ACL group, eight athletes injured their dominant leg and 13 athletes their nondominant leg. To reduce the possible influence of leg dominance in the comparisons between the ACL group and the control group, we included the same amount

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of dominant and nondominant legs in the control group (randomly selected).

Statistical analysis. To assess whether the ACL group and control group respond differently to fatigue, we used a two-way ANOVA with a mixed-model design (main effects for group and fatigue) to compare 1) the injured leg against the control group and 2) the uninjured leg against the control group. Furthermore, a two-way repeated-measures ANOVA (main effects for leg and fatigue) was used to compare 3) the injured leg against the contralateral, uninjured leg of the ACL group. We compared RPE scores (RPE-B and RPE-L), performance outcomes (e.g., jump distance of single-leg hop for distance, jump height of vertical hops with rotation), landing kinematics, kinetics and muscle activation between groups/legs, and prefatigue and postfatigue. To avoid unjustified reduction of the kinematic, kinetic, and muscle activation data to discrete values, such as mean or peak values, we used one-dimensional Statistical Parametric Mapping (SPM1D version M.0.4.5, www.spm1d.org) for our statistical analysis. SPM1D calculates test statistics at each time node, but avoids the problem of multiple comparisons by modeling the behavior of random time-varying signals for inference calculations (26). For all parameters that had as significant interaction effect, we conducted *post hoc* tests (SPM *t*-tests). A Bonferroni correction was used to correct for multiple testing ($\alpha = 0.05/4$). To determine the magnitude of significant differences of the *post hoc* tests, Cohen's *d* effect size (ES) was calculated at every time point and subsequently a mean ES was calculated for the time period that had a significant difference between groups. Differences with an effect size larger than 0.5 were of interest because they represent a moderate change.

RESULTS

RPE Scores

The ANOVA showed no interaction effect and no main effect for group for the RPE-B or RPE-L scores. However as expected, a main effect for fatigue was found, with higher RPE-L and RPE-B scores postfatigue compared with prefatigue (see Table 2).

Performance Outcomes

The ANOVA showed no interaction effect and no main effect for group for the jump performance. However, a main effect for fatigue was found for the jump distance of the single-leg hop and for the jump height of the vertical hop with 90° of lateral rotation (see Table 2).

Neuromuscular and Biomechanical Parameters

Interaction effects. Table 3 shows the interaction effects of the ANOVA analyses performed to compare 1) the ACL injured leg against the control group; 2) the injured leg against the contralateral, uninjured leg; and 3) the uninjured leg against the control group. Only a few interaction effects were found.

We found an interaction effect for the thorax flexion angles during the vertical hop with 90° of medial rotation when comparing the ACL injured leg to the control group (from 0 to 318 ms post-IC, $P < 0.001$) and comparing the ACL injured leg to the uninjured leg (from 64 to 150 ms post-IC, $P = 0.0023$ and from 201 to 369 ms post-IC, $P = 0.0025$). *Post hoc* analysis showed that thorax flexion angles are increased postfatigue compared with prefatigue in the ACL injured legs but not in the contralateral uninjured legs and even decreased in the control group, this resulted in significantly larger pelvis-thorax flexion angles in the ACL-injured leg compared with the control group around peak loading (72–106 ms post-IC, $P = 0.012$, ES = 0.55) (Fig. 1).

Furthermore, an interaction effect for the knee abduction moment was found during the lateral hop (43–53 ms post-IC, $P = 0.044$) when comparing the ACL injured legs compared with the control group. *Post hoc* analysis showed that knee abduction moments were only increased in the ACL injured legs and not in the control group postfatigue compared with prefatigue at peak loading (30–51 ms post-IC, $P = 0.0069$, ES = 0.57) (Fig. 2). In addition, a significant higher knee abduction moment was found in the control group postfatigue compared with prefatigue at the end of the landing phase (435–500 ms, $P < 0.001$, ES = 0.59) (Fig. 2). Because this finding is only at the end of the landing phase and not during the shock absorption, this seems less relevant for sustaining an ACL injury and will, thus, not be further discussed.

Fatigue effects. Table S1 (Supplemental Digital Content 1, overview of the main effects for fatigue, http://links.lww.com/MSS/B750) shows the main effects for fatigue of the ANOVA analyses performed to compare 1) the ACL injured leg against the control group; 2) the injured leg against the contralateral, uninjured leg; and 3) the uninjured leg against the control group. Several main effects for fatigue were found, especially in sagittal plane joint angles and moments.

Smaller hip, knee, and ankle flexion angles were found postfatigue compared with prefatigue during several tasks. Furthermore, knee abduction angles were increased postfatigue compared with prefatigue during the single-leg hop for distance.

Furthermore, thorax flexion moments were increased whereas hip and knee flexion moments were decreased postfatigue compared with prefatigue. In addition, knee abduction moments were increased postfatigue compared with prefatigue during the single-leg hop, lateral hop and medial hop.

Finally, some fatigue effects were found for muscle activation. We refer to Table S1 (Supplemental Digital Content 1, overview of the main effects for fatigue, http://links.lww.com/MSS/B750) for the detailed results as these effects were very diverse depending on which groups were included in the ANOVA.

Group effects. Table S2 (Supplemental Digital Content 2, overview of the main effects for group, http://links.lww.com/MSS/B751) shows the main effects for group of the ANOVA analyses performed to compare 1) the ACL injured leg against the control group; 2) the injured leg against the contralateral, uninjured leg; and 3) the uninjured leg against

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TABLE 2. ANOVA analyses for RPE scores and jump performance.

	ACL		Control				
	Prefatigue	Postfatigue	Prefatigue	Postfatigue	Group Effect	Fatigue Effect	Interaction Effect
RPE-B	9.43 ± 1.81 Very light	16.71 ± 2.19 Very hard	8.33 ± 2.32 Very light	16.05 ± 1.59 Very hard	$F = 2.789$ $P = 0.10$	$F = 466.50$ $P < 0.001$[a]	$F = 0.381$ $P = 0.54$
RPE-L	10.33 ± 1.39 Light	15.57 ± 2.22 Hard	9.43 ± 2.13 Very light	14.86 ± 2.36 Hard	$F = 1.959$ $P = 0.17$	$F = 320.82$ $P < 0.001$[a]	$F = 0.102$ $P = 0.75$
Jump distance SLH (m)	1.87 ± 0.26	1.79 ± 0.25	1.87 ± 0.32	1.77 ± 0.28	$F = 0.407$ $P = 0.66$	$F = 77.41$ $P < 0.001$[a]	$F = 0.485$ $P = 0.62$
Jump height 90 M (cm)	8.03 ± 1.70	7.70 ± 1.45	8.02 ± 2.14	7.51 ± 2.69	$F = 0.195$ $P = 0.82$	$F = 3.162$ $P = 0.080$	$F = 0.154$ $P = 0.85$
Jump height 90 L (cm)	8.01 ± 1.52	7.54 ± 1.41	8.28 ± 1.34	7.93 ± 1.14	$F = 0.657$ $P = 0.52$	$F = 9.565$ $P = 0.003$[a]	$F = 1.681$ $P = 0.19$

[a]Significant effect.
SLH, single-leg hop for distance; 90 M, vertical hop with 90° medial rotation; 90 L, vertical hop with 90° lateral rotation.

the control group. Only a few group effects were found. Increased hip adduction angles and decreased hip adduction moments were found in the ACLR legs compared with the control group, respectively during the vertical hop with 90° lateral rotation (324–500 ms, $P = 0.0053$) and the lateral hop (94–114 ms post-IC, $P = 0.033$). Finally an increased MH and LH activation was found in the ACL group compared to the control group (during all tasks) and compared to the contralateral leg (MH during all tasks; LH during lateral hop) (see Fig. S3, Supplemental Digital Content 3, which illustrates the group effect of MH activation, http://links.lww.com/MSS/B752).

DISCUSSION

This study showed that overall ACLR athletes and uninjured athletes have similar biomechanical and neuromuscular responses to fatigue induced by a match-play simulation protocol. The fatigue protocol mainly induced alterations in sagittal plane kinematics and kinetics. Across different jump modalities, most alterations were not more pronounced in the ACLR group compared to the control group (e.g., only a few interaction effects) and therefore our hypothesis that ACLR athletes might be more vulnerable to fatigue could not be confirmed. For 2 biomechanical parameters however, we did find an interaction effect in specific tasks, suggesting still that in some tasks landing deficits in ACLR athletes may become clearer under fatigued evaluation.

First of all, our results showed that fatigue induced an increased knee abduction moment in the ACLR injured legs, but not in the control group during the peak loading phase of the lateral hop. Since high knee abduction moments are associated with increased knee (re-)injury risk (27–29), one might suggest that fatigue predisposes ACLR athletes more for subsequent knee injuries than uninjured athletes because of the larger knee abduction moments. Interestingly, we found opposite results prefatigue with the ACLR athletes showing lower

TABLE 3. Overview of interaction effects with fatigue for the three ANOVA analyses comparing 1) the ACL injured leg against the control group; 2) the injured leg against the contralateral, uninjured leg; and 3) the uninjured leg against the control group.

	ACL Injured Leg—Control				ACL Injured Leg—ACL Uninjured Leg				ACL Uninjured Leg—Control			
	Task	P	Effect	Time	Task	P	Effect	Time	Task	P	Effect	Time
Joint Angles												
Thorax flexion	90 M	$P < 0.001$	ACL post↑ Control post↓	0–318 ms	90 M	$P = 0.023$ $P = 0.0025$	Inj post↑ Uninj post=	64–150 ms 201–369 ms		NS	—	—
Hip flexion		NS	—	—		NS	—	—		NS	—	—
Hip adduction		NS	—	—		NS	—	—		NS	—	—
Knee flexion		NS	—	—		NS	—	—		NS	—	—
Knee abduction		NS	—	—		NS	—	—		NS	—	—
Ankle dorsiflexion		NS	—	—		NS	—	—		NS	—	—
Joint moments												
Thorax flexion		NS	—	—		NS	—	—		NS	—	—
Hip flexion		NS	—	—		NS	—	—		NS	—	—
Hip adduction		NS	—	—		NS	—	—		NS	—	—
Knee flexion		NS	—	—		NS	—	—		NS	—	—
Knee abduction	LH	$P = 0.044$	ACL post↑ Control post=	43–54 ms		NS	—	—		NS	—	—
Ankle dorsiflexion												
Muscle activation												
VM		NS	—	—		NS	—	—		NS	—	—
VL		NS	—	—		NS	—	—		NS	—	—
HM		NS	—	—		NS	—	—		NS	—	—
HL		NS	—	—		NS	—	—		NS	—	—
GM		NS	—	—		NS	—	—		NS	—	—
GL		NS	—	—		NS	—	—		NS	—	—
GlutMed		NS	—	—		NS	—	—		NS	—	—

The interaction effect is described in the column "Effect" (for example: ACL post↑ means increased postfatigue values in the ACL group). NS means that in none of the tasks an interaction effect was found.
NS, not significant in any of the tasks; LH, lateral hop; 90 M, vertical hop with 90° medial rotation; Control, control group; Inj, ACL injured leg; Uninj, ACL uninjured leg. VL, vastus lateralis; VM, vastus medialis; GlutMed, gluteus medius.

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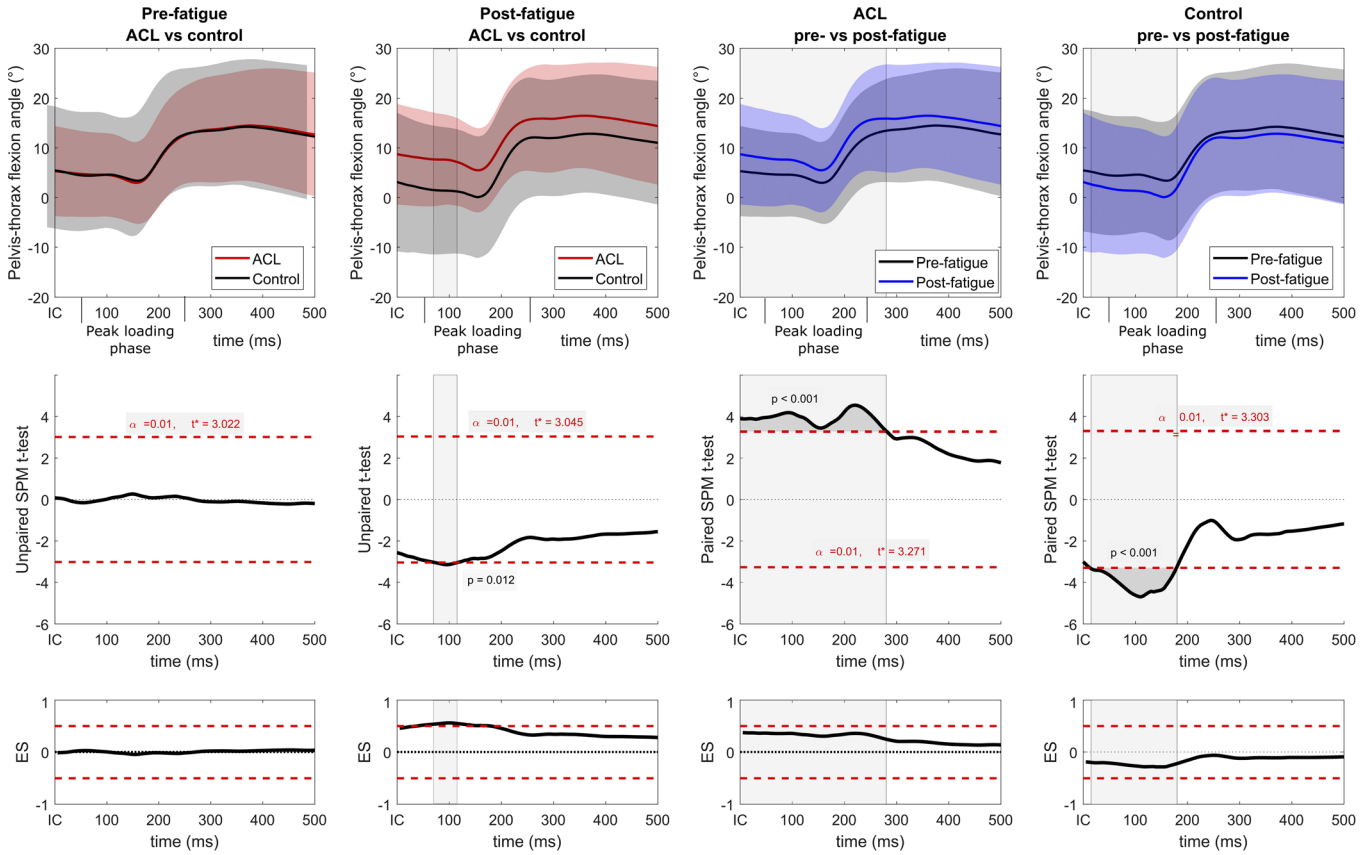

FIGURE 1—*Post hoc* analyses for pelvis-thorax flexion angle during the vertical hop with 90° medial rotation. Upper row: average pelvis-thorax flexion angle in the ACL group (*red*), control group (*black*), prefatigue (*black*), postfatigue (*blue*) during the landing phase (from IC until 500 ms after IC). Standard deviation clouds are represented by the shaded zones. Middle row: SPM output of the *t*-tests. If the *t*-curve (*black line*) exceeds the critical threshold (*horizontal red dashed line*) significant differences were found between groups. The periods in which significant differences were found, are highlighted in light gray on the corresponding graphs. Lower row: the effect size (ES) (*black line*) for the differences between groups. Two *red dashed lines* represent a moderate ES (ES = 0.5 and ES = −0.5, respectively).

knee abduction moments than the control group (see Fig. 2). This might indicate that ACLR athletes try to protect their knee prefatigue, but fail to reduce knee abduction moments once they are fatigued. The increased postfatigue knee abduction moments were only apparent during the performance of the lateral hop which might imply that the ACLR athletes had enough reserve to deal with fatigue during the other tasks and that task-demands might be decisive in whether enhanced deficits due to fatigue are apparent or not. Previous studies showed that lateral jump-landings require more frontal plane stability and result in larger peak knee valgus angles (30) compared with forward jump landings. Also in our study, the lateral hop seemed the most demanding task in the frontal plane as the highest average knee abduction moments were reached during this task.

Another interaction effect was found during the vertical hop with 90° medial rotation. Postfatigue thorax flexion angles were increased in the ACL injured legs, decreased in the control group, and did not change in the uninjured legs compared with prefatigue angles. Increasing the thorax flexion angle during peak loading may represent a compensatory strategy to reduce the peak loads on the knee joint by increasing the demands on the hip joint (31) as flexion of the trunk repositions the center of mass anteriorly and thus reduces sagittal knee joint moment arms and increases hip joint moment arms (32). This can in the first place be a protective strategy as high external knee flexion moments are associated with increased anterior tibial shear forces (33), increased ACL loading (34), and even increased primary ACL injury risk (35). On the other hand it can be a strategy to reduce eccentric work of the quadriceps with the aim to avoid knee collapse because of (fatigue) induced quadriceps weakness (36). We performed strength measurements in the beginning of the test session and found a significant quadriceps-strength deficit in the injured leg compared with the contralateral leg (injured: 2.56 ± 0.49 N·m·kg^{-1}, uninjured 2.75 ± 0.38 N·m·kg^{-1}, $P = 0.026$, see Table 1). It is expected that fatigue induces a quadriceps strength decrease (36). If fatigue has similar effects on the already weaker quadriceps in the injured leg of ACLR athletes, their capacity to create extension torque will further decrease postfatigue. Increasing trunk flexion and thus shifting demands from the knee toward the hip might help them to unload their fatigued quadriceps. Why this compensation was only seen in the vertical hop with 90° medial rotation is not yet clear.

The fact that only two interaction effects were found suggests that in general, biomechanical and neuromuscular responses to

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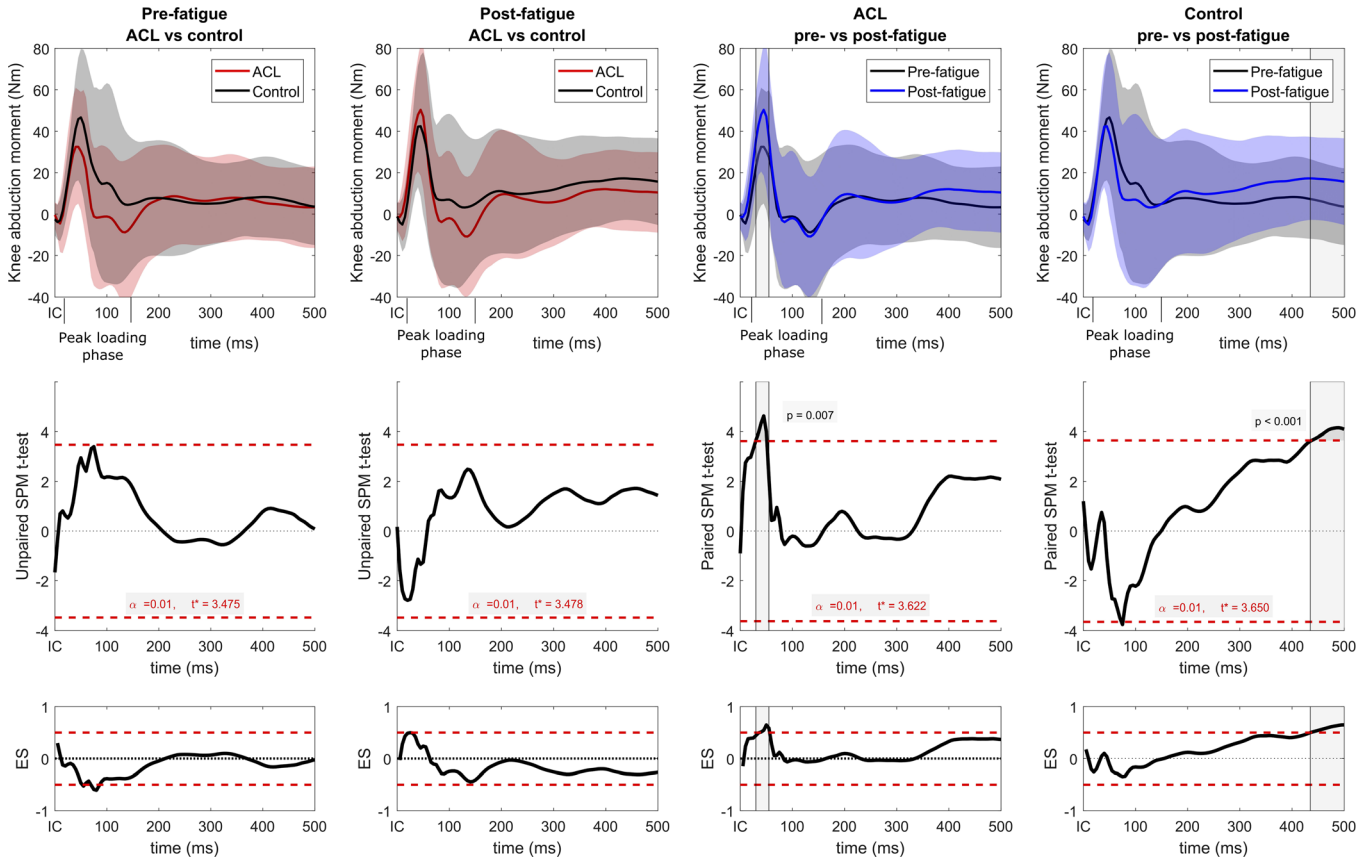

FIGURE 2—*Post hoc* analyses for knee abduction moment during the lateral hop. Upper row: average knee abduction moment in the ACL group (*red*), control group (*black*), prefatigue (*black*), postfatigue (*blue*) during the landing phase (from IC until 500 ms after IC). Standard deviation clouds are represented by the shaded zones. Middle row: SPM output of the *t*-tests. If the *t*-curve (*black line*) exceeds the critical threshold (*horizontal red dashed line*) significant differences were found between groups. The time periods in which significant differences were found, are highlighted in *light gray* on the corresponding graphs. Lower row: the effect size (ES) (*black line*) for the differences between groups. Two *red dashed lines* represent a moderate ES (ES = 0.5 and ES = −0.5, respectively).

fatigue are similar between ACLR athletes and uninjured athletes, which is in line with previous studies assessing fatigue effects (6,14,16). We found that both groups have a more extended landing pattern postfatigue compared to prefatigue, characterized by reduced ankle, knee and hip flexion angles, reduced external hip and knee flexion moments and increased thorax flexion moments. As this extended pattern was found across all tasks, it is very likely that this will also appear during sport-specific tasks on the field. These postfatigue kinematic and kinetic alterations might be an adaptive strategy to rely more on passive restraints and as such avoiding excessive eccentric quadriceps work to ensure a successful landing without collapse because of muscle fatigue (37). However, since a more extended pattern is linked with (re-)injury risk (12,35) we can conclude that this adaptation increases injury risk in both ACLR and uninjured athletes. It might thus be worthwhile for future studies to investigate whether large reduction in postfatigue quadriceps strength are related with more extended landing patterns.

We also found a few group effects. The increased MH and LH activation in the ACL injured legs compared with the control group and compared with the uninjured legs was the most important difference between groups. This seems a protective strategy as the hamstrings have a posterior line of pull in a flexed knee and might thus act as an ACL agonist, counteracting high anterior tibial shear forces (38). Furthermore, the ACLR athletes showed increased hip adduction angles and decreased hip adduction moments compared with the control group which might suggest hip abductor weakness (39).

This study comes with some limitations: the single-leg hop for distance and the vertical hops with 90° of rotation are maximal performance tests. As mentioned in the results section (Table 2), subjects jumped less far/high postfatigue, which probably also influenced postfatigue kinematics, kinetics and muscle activation. So far, there is no evidence yet if and how jump distance and jump height are related with alterations in kinematics and/or kinetics, and therefore, we were not able to correct for this decrease in performance. Nevertheless, large influences are not expected as the medial and lateral hop, two tasks that had equal jump height and distance before and after fatigue, showed similar alterations in postfatigue landing kinematics and kinetics as the tasks that had a decrease in performance. Furthermore the change in jump height was only very small (with effect sizes between 0.19 and 0.31) and thus of less clinical importance. A second limitation is the fact that perception of fatigue (measured with RPE scores) varied widely

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between the subjects, indicating that the same fatigue protocol (with fixed demands) induced different levels of fatigue between subjects, depending on their physical fitness levels, and psychosocial factors and recovery/sleep (11). The use of such a fixed-demand protocol instead of a protocol that induce a standardized level of fatigue in all subjects (e.g., complete exhaustion) has the advantage that it represents what happens in real match play, with fitter players typically experience less fatigue than players who have a poor fitness level. However, one should be aware that false negative outcomes are more likely as some subjects only reported very low RPE scores (e.g., RPE-B of 11, "fairly light") (20). Finally, fatigue effects were simultated with a short-term fatigue protocol and thus not perfectly representative of fatigue effects because of long-duration match play. Although a previous study of Bossuyt et al. (21) confirmed that the SAFT5 has a similar overall pattern in heart rate and RPE values as actual match play and was also able to induce neuromuscular alterations (e.g., strength reductions), probably different physiological processes will be triggered in long-duration match play. Furthermore, the SAFT5 protocol mainly included vertical and sagittal plane movements, which might explain why most fatigue effects were found in the sagittal plane.

As stated in the Introduction, we hypothesized that previous studies assessing fatigue in ACLR athletes did not find increased vulnerability for fatigue in ACLR athletes because the tasks assessed were insufficiently demanding. Although the tasks of our study were more challenging (e.g., maximal performance, directional changes), we only found limited differences in biomechanical response to fatigue between groups. As explained earlier, the lateral hop task was the most demanding task in the frontal plane, and was the only task that revealed a larger increase in postfatigue knee abduction moment in the ACL injured legs. Possibly, this was the only task where the ACLR athletes do not have enough reserve to cope with fatigue. Based on these findings, we advise to not only use forward (e.g., single-leg hop for distance) or vertical tasks (e.g., drop vertical jump) in RTS test batteries but also include frontal plane tasks, such as the lateral hop, especially because most ACL injury mechanisms involve a lateral component as well (40). Because growing evidence exists that combining fatigue and decision making has an integrative effect on landing kinematics and kinetics in uninjured athletes (11,41), future studies should assess how ACLR athletes respond when fatigue and decision making is combined. Such open environments further increase the demands and as this is what athletes can experience during match play one may wish to evaluate the existence of biomechanical and neuromuscular deficits also under such circumstances.

CONCLUSIONS

In conclusion, this study found that ACLR athletes and uninjured athletes show in general similar biomechanical and neuromuscular responses to fatigue. However, some interaction effects were found, suggesting that fatigue enhanced landing deficits in ACLR athletes may become more exposed in certain tasks than others. For example, during the landing of the lateral hop, only the ACLR leg showed increased postfatigue knee abduction moments compared with prefatigue, whereas knee abduction moments in the uninjured leg and control group were not affected by fatigue. Therefore, we advise to not only use forward or vertical tasks in RTS test batteries, but also include frontal plane tasks, such as the lateral hop. Future studies should further assess how task demands influence ACLR athletes' ability to cope with fatigue effects.

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